

WACO, TX, USA

WMO: 722560

Lat: 31.619N

Lon: 97.228W

Elev: 152

StdP: 99.51

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13959

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.2	-2.3	-11.3	1.5	0.0	-9.1	1.8	2.8	12.2	10.5	11.1	11.7	3.9	0	0.508

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.9	38.5	23.4	37.4	23.7	36.3	23.8	25.8	32.8	25.5	32.4	25.1	31.9	4.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	19.5	27.7	23.8	19.0	27.5	23.4	18.5	27.4	80.4	33.0	78.6	32.2	76.9	31.9	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.7	9.3	8.5	-7.5	40.4	2.2	1.8	-9.0	41.6	-10.3	42.7	-11.5	43.7	-13.1	44.9		
			-8.4	26.7	2.0	0.9	-9.9	27.4	-11.1	27.9	-12.2	28.4	-13.7	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.8	8.9	11.1	15.0	19.3	23.8	27.9	29.8	29.9	26.3	20.5	14.5	9.7
	DBStd	8.63	5.45	5.74	5.10	4.19	3.52	2.43	1.98	2.26	3.44	4.55	5.21	5.27
	HDD10.0	241	85	50	16	1	0	0	0	0	0	1	18	71
	HDD18.3	1113	293	209	127	39	4	0	0	0	2	31	136	272
	CDD10.0	3808	53	82	170	279	428	537	614	617	488	327	153	62
	CDD18.3	1637	3	7	22	67	174	287	356	359	239	98	20	4
	CDH23.3	19045	19	53	148	526	1658	3417	4726	4775	2664	909	127	22
	CDH26.7	9660	2	9	22	129	651	1738	2677	2753	1329	328	20	2
Wind	WSAvg	4.3	4.3	4.6	4.8	5.0	4.6	4.4	4.2	3.8	3.4	3.9	4.1	4.1
Precipitation	PrecAvg	860	53	59	69	80	113	83	48	49	79	100	66	62
	PrecMax	1511	155	201	248	238	381	306	218	262	241	386	247	249
	PrecMin	497	1	2	2	3	14	5	0	0	4	0	2	4
	PrecStd	202	39	39	46	56	82	65	53	56	59	86	49	53
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.8	27.8	29.1	32.4	35.6	38.8	39.7	40.3	38.6	34.0	29.0	26.0
		MCWB	16.0	16.7	18.8	20.8	23.1	23.6	23.4	23.0	22.2	22.0	20.0	17.7
	2%	DB	22.8	24.8	26.7	29.7	33.5	36.8	38.3	38.7	36.3	32.2	26.5	23.0
		MCWB	15.8	16.6	18.5	20.6	23.3	23.8	23.5	23.5	23.2	21.9	19.3	17.3
	5%	DB	20.5	22.4	24.9	28.0	31.9	35.3	37.1	37.6	34.7	30.3	24.6	20.9
		MCWB	15.0	16.2	18.0	20.2	23.0	24.0	23.8	23.8	23.0	20.9	18.1	16.0
	10%	DB	18.3	20.4	23.0	26.4	30.3	33.9	35.9	36.4	33.2	28.3	22.7	18.8
		MCWB	13.9	15.9	17.3	19.5	22.6	23.9	23.9	23.8	22.7	20.1	18.0	14.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.4	20.4	21.4	23.9	25.9	26.7	26.1	26.2	25.8	24.7	22.5	20.3
		MCDB	22.1	23.7	25.7	28.8	31.3	33.8	33.4	33.9	31.9	29.7	25.9	23.0
	2%	WB	18.1	19.1	20.4	22.7	24.9	25.8	25.6	25.6	25.2	23.8	21.1	18.9
		MCDB	20.6	21.9	24.2	27.4	30.2	32.6	33.1	33.2	31.1	28.4	24.1	21.5
	5%	WB	16.7	18.0	19.6	21.7	24.1	25.3	25.2	25.2	24.6	23.0	20.1	17.4
		MCDB	19.3	20.9	22.9	26.0	29.3	31.6	32.7	32.7	30.4	27.4	23.1	20.2
	10%	WB	14.7	16.6	18.6	20.9	23.5	24.8	24.9	24.8	24.1	22.3	18.8	15.5
		MCDB	17.5	19.8	21.8	24.7	28.6	30.9	32.0	32.0	29.7	26.2	22.1	18.1
Mean Daily Temperature Range	5% DB	MDBR	12.2	11.8	11.8	11.9	11.1	11.0	11.6	11.9	12.2	12.5	11.8	11.6
		MCDBR	15.6	14.5	13.1	12.7	12.1	12.4	12.9	13.2	13.3	13.7	13.1	14.0
		MCWBR	9.4	8.6	7.0	6.2	4.5	3.1	2.7	2.9	3.8	5.7	7.1	8.7
	5% WB	MCDBR	11.7	11.9	10.6	10.1	10.4	10.4	11.1	11.5	10.8	10.5	10.3	11.4
		MCWBR	8.7	8.6	6.9	5.9	4.7	3.5	3.0	3.0	3.6	5.2	7.4	8.6
Clear-Sky Solar Irradiance	taub	0.317	0.338	0.359	0.400	0.429	0.434	0.445	0.439	0.403	0.370	0.348	0.325	
	taud	2.505	2.453	2.410	2.317	2.296	2.318	2.322	2.323	2.371	2.447	2.456	2.487	
	Ebn at Noon	910	921	922	892	864	855	844	846	866	873	865	880	
	Edh at Noon	87	101	113	129	133	130	129	127	116	100	91	85	
All-Sky Solar Radiation	RadAvg	2.95	3.42	4.48	5.43	5.79	6.57	6.72	6.21	5.21	4.16	3.11	2.59	
	RadStd	0.30	0.58	0.38	0.47	0.51	0.54	0.47	0.38	0.43	0.61	0.38	0.35	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	-0.37	N/A	N/A	N/A	N/A
(47)	Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page